

Residual Trust Boundaries in Satellite Cyber Recovery: Temporal Evidence, Producer Composition, and Artifact Assurance

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Abstract—Trusted cyber recovery for satellite systems can depend on policy-visible evidence whose authenticity, freshness, composition, or provenance remains acceptable even when research-only authorization or correctness differs. This paper reports three separately frozen deterministic studies of recovery qualification without pooling their populations. Study 3 evaluates 1,380 temporal trajectories and shows that affected post-signature-manipulated records are rejected, whereas false but validly signed evidence from a compromised trusted producer can persist or recur across continuous and synthetic intermittent-contact schedules; the design separately identifies ordinary pre-onset cache exposure. Study 4 evaluates 4,608 exact observations across 18 vote and synthetic provenance-domain rules. Provenance constraints delay systematic unsafe qualification for selected thresholds without always changing first failure, while some constraints produce earlier false-conservative rejection under benign producer loss and others produce no threshold effect. Study 6 evaluates 420 exact artifact-assurance observations. Composed assurance gates reduce the prespecified incorrect states that remain qualified from four of five under signature-only checking to one of five under the six-signal composite gate, while increasing sensitivity to benign assurance-signal unavailability. Across the three finite models, stronger trust composition closes specified failure pathways but does not automatically make policy-visible evidence equivalent to hidden or objective truth. The results characterize residual trust boundaries rather than a globally best recovery policy, producer-composition rule, or artifact gate.

Index Terms—Aerospace cybersecurity, cyber recovery, evidence qualification, trust management, software supply-chain assurance.

I. INTRODUCTION

Cyber recovery in satellite systems can require trust decisions under communication gaps, constrained postlaunch access, and mission-continuity pressure [1]. Recovery is therefore not only a restoration problem but also a qualification problem: before a policy permits a recovery path, it must decide whether the available runtime evidence, the producers supplying that evidence, and the recovery artifact are sufficiently trustworthy for the modeled decision.

Prior work already establishes the main mechanisms used here. Satellite research examines internal trust boundaries and testable cyber-resilience requirements [2], [3], while

SPARTA describes integrity-protected trusted recovery baselines [4]. RATS distinguishes evidence from appraisal and treats freshness as an explicit concern [5], with current work also considering multiple-Verifier composition [6]. Quorum systems formalize trust and failure assumptions [7], [8], and recent satellite trusted-execution work uses Byzantine-tolerant endorsement quorums [9]. Software-supply-chain systems provide provenance, signed metadata, hashes, controlled build processes, and source assurance [10], [11], [12], while SLSA also documents limits involving intentionally malicious producers [13]. This paper does not claim novelty for those mechanisms.

The narrower question is: **what residual trust boundary remains when a recovery-qualification policy can satisfy its visible checks while a relevant research-only authorization or correctness state differs?** Runtime evidence can be authentic and fresh but semantically false if a trusted producer is compromised. Multiple producers can reduce dependence on one source, but the boundary then depends on vote count, provenance assumptions, and producer availability. Artifact assurance can close several integrity and process failures while still depending on a higher-level assumption about approved source correctness.

Three separately frozen deterministic studies examine those boundaries without forming one integrated experiment or pooled population. Study 3 evaluates 1,380 temporal trajectories under continuous and synthetic intermittent contact. Study 4 evaluates 4,608 exact rule-by-subset observations across 18 vote and synthetic provenance-domain rules. Study 6 evaluates 420 exact observations across six artifact states, six assurance gates, and benign assurance-signal unavailability. Each study retains its own unit, interventions, endpoints, and model boundary.

The research questions are:

RQ1: Under the frozen continuous and intermittent-contact schedules, how do truthful evidence, post-signature modification, and false but validly signed claims from a compromised trusted producer affect the duration and recurrence of false recovery qualification across the Study-3 policy semantics?

RQ2: When recovery authorization is supported by multiple modeled evidence producers, how do absolute vote thresh-

olds and synthetic provenance-domain requirements change the first and systematic failure boundaries for unsafe qualification under producer compromise and false-conservative qualification under benign producer loss?

RQ3: When the recovery artifact is subjected to progressively composed assurance requirements, which prespecified incorrect artifact states remain qualified, and what benign qualification loss results when required assurance signals become unavailable?

A systems-level synthesis then asks how these independently measured boundaries relate across temporal runtime evidence, producer composition, and recovery-artifact assurance. The synthesis is qualitative and mechanism based; it does not estimate a common treatment effect, pooled success rate, or end-to-end recovery probability.

The paper makes four bounded contributions:

1. **Temporal qualification boundary.** Study 3 separates a short truthful pre-onset cache boundary from false qualification caused by a compromised trusted producer that continues to generate fresh and validly signed false evidence, while the post-signature-manipulation control distinguishes semantic producer failure from invalidly signed alteration.
2. **Producer-composition boundary.** Study 4 provides an exhaustive first-versus-systematic failure map across 18 vote and provenance-domain rules, including conditional benefits, benign-loss costs, and null provenance effects.
3. **Artifact-assurance boundary.** Study 6 maps the incorrect artifact states that survive progressively composed assurance gates and the corresponding rejection of the objectively correct baseline under benign signal loss.
4. **Residual-boundary synthesis.** Across the three studies, stronger evidence composition closes or narrows specified failure pathways without automatically making policy-visible evidence equivalent to hidden or objective truth.

The interpretation remains model bounded. Only Study 3 models contact, and logical time is not operational spacecraft time. Study 4's provenance domains are synthetic, and Study 6 is a Boolean assurance model rather than a real supply-chain experiment. The studies do not measure flight safety, mission availability, RF performance, computing cost, or operational recovery probability; Section VIII states the complete validity and aerospace interpretation limits.

Section II positions the work against prior art. Section III defines the common qualification abstraction while preserving study separation. Sections IV through VI present the three studies, Section VII synthesizes their residual boundaries, Section VIII addresses validity and future evaluation, and Section IX concludes.

II. RELATED WORK AND SCIENTIFIC POSITIONING

A. Space Cybersecurity and Recovery Context

Spacecraft cybersecurity is shaped by operating constraints that can make evidence collection, human intervention, and recovery qualitatively different from continuously connected terrestrial systems. Thummala, Rice, and Falco identify communication gaps, permanent loss of physical access after launch, tight subsystem coupling, and mission-continuity requirements as interacting characteristics of the space-security problem [1]. Their argument is important here because it motivates recovery decisions that may need to be made from incomplete or intermittently refreshed evidence. It does not, however, imply that any particular contact schedule used in this paper is an orbital or ground-station visibility model.

Recent work also emphasizes that trust boundaries inside satellite flight software can be weak even when components operate through legitimate interfaces. Vanlyssel et al. analyze cFS and other modular flight-software frameworks and show how a compromised onboard component can abuse authority that is architecturally available to trusted peers [2]. Curbo and Falco similarly argue for testable, secure-by-design flight-software requirements that make cyber-resilience properties explicit rather than treating security as an untestable global property [3]. These studies reinforce a systems question that is central to cyber recovery: which evidence and authority relationships should a recovery mechanism trust when a component, producer, or software artifact may remain syntactically valid while its semantic trustworthiness has changed?

Space-specific guidance already recognizes the need for controlled recovery baselines. The Aerospace Corporation's SPARTA material describes cyber-safe recovery from an integrity-protected and validated software and configuration baseline, together with authenticated and integrity-verified maintenance of recoverable trusted versions [4]. The present work therefore does not claim that trusted recovery baselines, cyber-safe mode, or integrity-protected recovery are new concepts. Instead, it examines narrower qualification questions that arise before a policy treats runtime evidence, producer agreement, or a recovery artifact as sufficiently trustworthy.

B. Evidence Appraisal, Freshness, and Policy-Visible Trust

The IETF Remote Attestation procedureS architecture provides a mature vocabulary for evidence-based trust decisions. RFC 9334 distinguishes an Attester that produces Evidence, a Verifier that appraises Evidence under an appraisal policy, and a Relying Party that consumes Attestation Results to make application-specific decisions [5]. It also treats freshness as an explicit appraisal concern. Freshness can bound how old evidence is relative to the policy's tolerance, but it cannot guarantee instantaneous synchronization with a state that may change immediately after evidence is generated [5].

That distinction matters for Study 3. A record may be authentic and still fresh under the policy while no longer

matching the research-only authorization truth. Conversely, a compromised trusted producer can generate a false claim that remains both fresh and cryptographically valid. The contribution of Study 3 is therefore not the observation that freshness matters or that signatures authenticate origin. Its contribution is the exact temporal characterization of the remaining false-qualification exposure across the frozen onset, contact, persistence, and policy grid, including prespecified separation of ordinary pre-onset cache exposure from false evidence generated by the compromised trusted producer.

Current RATS work also considers complex attestation topologies. The 2026 RATS Working Group draft on multiple Verifiers describes hierarchical, cascaded, and hybrid patterns in which different Verifiers appraise portions of a composite system and coordinate partial Evidence or partial Attestation Results [6]. This is related to the broader problem of composing trust evidence, but it is not the mechanism evaluated by Study 4. Study 4 does not distribute appraisal across multiple Verifiers. It models seven evidence producers whose visible claims are evaluated by a recovery-qualification rule requiring an absolute vote threshold and, for selected rules, a minimum number of synthetic provenance domains. The distinction between evidence producers and evidence Verifiers is maintained throughout this paper.

C. Quorum Trust and Producer Composition

Quorum systems are established foundations for fault-tolerant distributed computing. Malkhi and Reiter formalized Byzantine quorum systems that preserve consistency and availability under arbitrary repository failures [7]. More recent work extends quorum reasoning to asymmetric or subjective trust assumptions in which participants can hold different fail-prone assumptions [8]. These results establish that quorum structure, fault assumptions, consistency, and availability tradeoffs are not new contributions of this paper.

Satellite-specific work further narrows the novelty boundary. Space Fabric uses a Byzantine-tolerant endorsement quorum of distributed ground stations and diversified secure elements as part of a satellite-enhanced trusted-execution and attestation architecture [9]. Thus, combining satellite systems with quorum endorsement or diversified trust anchors is also not, by itself, a novel contribution of Study 4.

Study 4 asks a more limited recovery-qualification question. Given a fixed registered set of seven modeled evidence producers assigned to three synthetic provenance domains, it exhaustively evaluates every producer subset under 18 prespecified total-vote and provenance-domain rules. The experiment records both the first affected-producer count at which a qualification failure becomes possible and the affected-producer count at which failure becomes systematic across all subsets of that size. It evaluates malicious producer compromise and benign producer unavailability in separate exhaustive blocks. The resulting contribution is an exact threshold map for this finite recovery-evidence model,

including conditions in which provenance diversity changes³ systematic failure without changing first failure, and conditions in which provenance requirements have no additional effect. The model is not a Byzantine consensus protocol, does not establish real organizational or hardware independence, and does not estimate compromise or outage probabilities.

D. Artifact Provenance and Software Supply-Chain Assurance

Software-supply-chain security already provides several mechanisms relevant to recovery-artifact qualification. *in-toto* uses cryptographically verifiable metadata to provide evidence about the sequence of steps used to produce software [10]. The Update Framework uses trusted roles, signed metadata, cryptographic hashes, expiration, versioning, and configurable signature thresholds to secure software update distribution and limit the effects of key compromise [11]. SLSA defines source and build assurance requirements intended to increase confidence that software was created through expected, auditable processes [12]. These mechanisms establish provenance, target binding, controlled build processes, and authenticated release metadata as prior art.

The current SLSA v1.2 threat model is especially relevant to the interpretation of Study 6. SLSA explicitly states that an intentionally malicious software producer cannot be directly mitigated by SLSA controls alone and that consumers need some independent basis for trusting the producer [13]. Therefore, Study 6's `APPROVED_BAD_SOURCE` state must not be presented as the discovery that provenance or process compliance cannot prove benevolent source intent. That limitation is already recognized in contemporary supply-chain assurance.

Study 6 instead provides an exact residual-state map for a frozen six-state, six-gate artifact model. It asks which prespecified incorrect recovery-artifact states remain qualified as signature, digest, provenance, reproduced-build, source-review, and approval signals are composed, and it separately enumerates the benign qualification loss produced when required assurance signals become unavailable. Equal aggregate counts are preserved when different gates leave different residual states. The resulting contribution is not a new provenance mechanism. It is a finite characterization of which modeled trust assumptions each gate closes and which assumption remains outside the gate's observability.

E. Scientific Positioning and Gap Addressed

The literature reviewed above establishes the main primitives that appear in this paper. Space cybersecurity already treats communication gaps, autonomy, continuity, trusted baselines, and internal trust boundaries as important concerns [1], [2], [3], [4]. RATS already formalizes Evidence, appraisal, Relying Parties, and freshness [5], [6]. Distributed-systems research already formalizes quorum trust and availability relationships [7], [8], and current satellite trust ar-

chitectures already use endorsement quorums [9]. Software-supply-chain research already provides provenance, signed update metadata, target binding, reproducible-process concepts, and explicit limits on what those controls can establish about producer intent [10], [11], [12], [13].

Accordingly, this paper does not claim novelty for any of those individual mechanisms. Its contribution is the exact characterization of three distinct residual recovery-qualification boundaries under separately frozen finite models. Study 3 isolates temporal evidence validity and trusted-producer semantics under a frozen continuous or intermittent-contact schedule. Study 4 isolates producer-count and synthetic provenance-domain composition under exhaustive compromise and benign-unavailability subsets. Study 6 isolates recovery-artifact assurance composition under prespecified incorrect states and exhaustive benign assurance-signal loss.

In the reviewed literature, we did not identify a directly matching spacecraft cyber-recovery study that reports these three residual qualification perspectives as separately frozen experiments while preserving their distinct populations and then synthesizes them without pooling. That observation is a literature-positioning statement, not a priority claim. The paper's systems contribution is therefore a bounded residual-trust interpretation: adding or composing trusted evidence can move a qualification boundary and close specified failure pathways, but the remaining boundary depends on what the policy can observe and what trust assumptions are left outside that observation set.

III. COMMON TRUST-QUALIFICATION FRAMEWORK AND STUDY SEPARATION

A. Policy-Visible Evidence and Research-Only Truth

The studies use different mechanisms but share one analytical structure. For study j , let E_j denote the policy-visible evidence and let

$$Q_j(E_j) \in \{0, 1\}$$

represent the frozen qualification decision. Let $T_j \in \{0, 1\}$ denote the corresponding research-only adjudication state, where 1 means that hidden authorization or objective artifact correctness supports qualification. T_j is used only for evaluation and is never supplied to Q_j .

The generic unsafe-qualification condition is

$$U_j = 1[Q_j(E_j) = 1 \wedge T_j = 0].$$

Where a study separately evaluates benign evidence loss, the complementary false-conservative condition is

$$C_j = 1[Q_j(E_j) = 0 \wedge T_j = 1].$$

These expressions are manuscript-level abstractions, not new experimental endpoints. Each study retains its frozen unit, endpoint definitions, state space, and reporting rules.

Here, **residual trust boundary** means the remaining modeled states in which visible evidence satisfies qualification while research-only adjudication is false, together with any study-specific boundary at which stronger visible-evidence requirements cause false-conservative rejection. The term is descriptive of the registered finite models, not a universal impossibility theorem.

B. Study-Specific Realizations

Table I maps the shared abstraction to the three studies without merging them.

In Study 3, Q_3 evaluates received evidence under frozen signature, freshness, policy, and contact semantics while T_3 is hidden authorization truth. The design distinguishes truthful cache-origin false qualification from false but validly signed evidence generated by a compromised trusted producer. Full timing, treatment, and policy details are given in Section IV.

In Study 4, Q_4 applies total-vote and synthetic provenance-domain requirements to seven signed producer claims, while T_4 is hidden authorization truth. Producer compromise and benign producer unavailability are evaluated in separate exhaustive blocks. Full producer assignments and rule definitions are given in Section V.

In Study 6, Q_6 composes visible artifact-assurance signals while T_6 is objective baseline correctness. Incorrect artifact states and benign assurance-signal unavailability are evaluated separately. Full state, signal, and gate definitions are given in Section VI.

Only Study 3 directly models intermittent contact. Study 4 producer unavailability and Study 6 assurance-signal unavailability are distinct constructs and are not contact loss or mission availability.

C. Exact Finite Populations and Non-Pooling

All three studies are deterministic finite experiments rather than random samples from an operational population. Their populations remain separate because their units, interventions, and outcomes differ: Study 3 uses trajectories, Study 4 uses rule-by-subset observations, and Study 6 uses artifact-state and assurance-unavailability observations. No pooled Paper-2 N , success rate, confidence interval, p-value, common effect size, or global policy ranking is defined.

Section VII therefore compares mechanisms and residual boundaries qualitatively rather than estimating a common treatment effect or end-to-end recovery probability.

D. Qualification Is Not Recovery Completion

The modeled endpoint is qualification, not completed spacecraft recovery. A permissive or qualified gate does not establish that a recovery action executed successfully, restored mission capability, or produced a safe operational state. Likewise, Study 4's availability terminology refers to

TABLE I
STUDY-SPECIFIC REALIZATIONS

Study	Qualification layer	Research-only adjudication	Principal policy-visible evidence	Frozen population	Contact model
Study 3, S3-K4E-001	Temporal runtime evidence	Hidden authorization truth	Signature validity, trusted source semantics, freshness, epoch/contact-dependent record availability, security signal	1,380 trajectories	Yes, K0 and synthetic K4
Study 4, S4-MPQ-001	Producer composition	Hidden authorization truth	Signed producer claims, total-vote threshold, synthetic provenance-domain count	4,608 exact rule-by-subset observations	No
Study 6, S6-SCTR-001	Recovery-artifact assurance	Objective baseline correctness	Signature, independent target digest, provenance, independent reproduced build, source-review attestation, release approval	420 exact observations	No

false-conservative qualification under benign producer loss, and Study 6's benign availability loss refers to rejection under missing assurance signals. Neither is mission availability. Section VIII provides the full construct and external-validity boundaries.

E. Frozen Provenance and Reproducibility

Each study is bound to a frozen design and repository execution. Independent same-repository audits reported zero mismatches for Study 3 trajectories and origin rules, Study 4 reconstructed observations and thresholds, and Study 6 frozen outputs. These controls support reproducibility of the reported finite experiments; they are not external empirical replication.

F. Cross-Study Interpretation Rule

The common synthesis is limited to observability. Stronger evidence composition can close specific modeled failure pathways, but qualification remains constrained by what the gate can observe and by trust assumptions that remain outside that observation set. Sections IV through VI establish those boundaries separately before Section VII compares them.

IV. TEMPORAL EVIDENCE QUALIFICATION UNDER INTERMITTENT CONTACT

A. Study Question and Design

Study 3 (S3-K4E-001) evaluates when runtime recovery evidence remains policy-qualified after research-only authorization has changed. It varies contact availability, evidence semantics, and whether an affected post-onset record occurs once or persists. The model is deterministic and uses logical time rather than wall-clock or flight time.

The horizon is 240 logical seconds in five-logical-second epochs, with five logical seconds of evidence validity. The onset grid contains 46 prespecified phases from 10 through 235 logical seconds in five-second increments. Hidden authorization is `true` before onset and `false` at and after onset; the post-onset security signal is `true`. Hidden authorization is never supplied to the selector.

K0 provides continuous modeled contact from 0 through 240. K4 provides synthetic windows [25, 35], [75, 90],

[145, 165], and [220, 240]. New records arrive only during modeled contact; otherwise the selector can use the latest received record subject to freshness. K4 is a deterministic contact treatment, not an orbital pass schedule or operational-access estimate.

The evidence treatments are:

1. V0: truthful evidence. A post-onset record reports authorization `false` with a valid signature.
2. V4: post-signature value manipulation. The signed post-onset value is changed from `false` to `true`, invalidating the signature.
3. V5: compromised trusted-producer evidence. The producer reports authorization `true` while hidden truth is `false`, and validly signs the false claim.

V4 and V5 are each evaluated as one-shot and persistent treatments; V0 remains truthful. The frozen policy semantics are `S2_B0_FAIL_CLOSED`, `S2_B2_RISK_THRESHOLD`, and `S2_S1_EVIDENCE_AWARE`. The identifiers reuse frozen selector semantics, but no Study-2 result is imported. Thirty cells crossed with 46 onset phases yield 1,380 trajectories and 67,620 epoch states. The trajectory is the study unit; epochs within a trajectory are repeated model states.

B. Endpoints and Origin Decomposition

Primary endpoints are `unsafe_permissive_epoch_rate`, `unsafe_qualified_epoch_rate`, `unsafe_qualified_exposure_s`, `unsafe_qualified_episode_count`, `protective_epoch_rate`, and `action_transition_count`.

This paper emphasizes `unsafe_qualified`: the gate is policy-visible qualified while hidden authorization is `false`. `unsafe_permissive` is only a selector or gate-entry action metric, not completed recovery.

False-qualified epochs must map to one of two prespecified origins. `PRE_ONSET_CACHE` is a truthful record received before onset that remains policy-fresh afterward; `V5_AFFECTED_RECORD` is a validly signed false record from the compromised trusted producer. This separates ordinary freshness lag from adversarial semantic falsity. Because the

frozen onset grid is complete, results are exact finite-grid summaries and paired phase differences, with no p-value gate, weighted policy score, or global policy rank.

C. Persistent False-but-Valid Evidence Under Continuous Contact

Under persistent V5/K0, both B0 and S1 were unsafe-qualified in all 46 onset trajectories, with mean exposure 122.5 logical seconds. B2 remained 0/46 with zero mean exposure in this frozen cell. The V5 records are validly signed; the mismatch exists because the trusted producer itself signs a claim that is false relative to hidden authorization. Signature validity therefore authenticates the modeled producer and record integrity without exposing producer-origin semantic falsity.

The B2 zero is structural to the frozen policy and treatment grid, not evidence of universal immunity or global superiority.

D. Persistent V5 Under the Frozen K4 Contact Schedule

Under persistent V5/K4, B0 remained unsafe-qualified in 46/46 trajectories with mean exposure 55.326 logical seconds and a 55.0-logical-second increment over truthful V0. S1 remained unsafe-qualified in 46/46 with mean exposure 49.022 logical seconds, equal to its V5-attributable increment. B2 remained 0/46. The B0-S1 mean difference was approximately 6.304 logical seconds, so the frozen S1 contact-aware restriction reduced exposure relative to B0 but did not eliminate the V5 boundary.

K4 mechanically limits record reception under the registered cache semantics; the result does not establish that intermittent contact improves security. Table II values are logical model time, not spacecraft response time, communication latency, operator latency, or ground-contact duration.

E. Post-Signature Manipulation and the Cryptographic Boundary

Under V4, post-signature modification invalidates the affected record's signature, and affected V4 records never qualify. Any B0/K4 false qualification in V4 cells comes instead from a previously received truthful record and is assigned to PRE_ONSET_CACHE, not to the manipulated record.

The V4-V5 contrast therefore supports two bounded claims: signature validation rejects the modeled post-signature alteration, while a valid signature alone does not establish semantic truth when the trusted producer signs the false claim. The experiment does not evaluate cryptanalysis, key extraction, or a real signing-system attack.

F. Freshness and the Truthful Cache Boundary

Under truthful V0/K4/B0, 3/46 onset trajectories contained unsafe qualification, with mean exposure 0.326 logical seconds, all attributed to PRE_ONSET_CACHE. S1 and B2

had no truthful-V0 false qualification under K4. The result is schedule-, epoch-, freshness-, and policy-specific and is not an estimate of spacecraft cache staleness. It demonstrates that ordinary freshness lag can produce false qualification without being conflated with V5 producer-origin false evidence, consistent with RFC 9334's freshness limitation [5].

G. One-Shot V5 and Temporal Persistence

One-shot V5/K0 produced five logical seconds of mean unsafe-qualified exposure for both B0 and S1 across all 46 onset phases. Under K4, all 46 B0 and S1 trajectories eventually received the one-shot compromised record; S1 retained five logical seconds of V5 exposure, while B0 also contains the separately identified cache boundary. One-shot and persistent V5 are cryptographically the same false-but-valid producer claim; persistence determines whether the claim appears once or is renewed after later contact.

H. Study-3 Residual Trust Boundary

Study 3 therefore separates two residual boundaries: brief truthful freshness lag and false but validly signed evidence from a compromised trusted producer. Contact-aware restrictions reduce selected K4 exposure but do not eliminate persistent V5 qualification for B0 or S1. These are exact properties of the frozen grid, not estimates of compromise prevalence, unsafe-recovery probability, operational mission risk, or real spacecraft timing.

V. MULTI-PRODUCER QUALIFICATION AND PROVENANCE-DOMAIN CONSTRAINTS

A. Study Question and Exact Population

Study 4 (S4-MPQ-001) evaluates how absolute vote count and synthetic provenance-domain requirements change recovery qualification under two separate conditions: malicious producer compromise and benign producer unavailability.

Seven producers P1-P7 are assigned to frozen domains $D1=\{P1, P2, P3\}$, $D2=\{P4, P5\}$, and $D3=\{P6, P7\}$. These are synthetic independence classes, not demonstrated organizational, hardware, software, supply-chain, or operator independence. A rule Qq_Dd requires q qualifying claims and d represented domains, with $q=1..7$ and $d=1..\min(3, q)$, yielding 18 rules. The denominator is always the registered seven-producer set.

In the safety block, hidden authorization is false; all producers are available; compromised producers emit validly signed authorization-true claims and honest producers emit false. `unsafe_qualified` is true when the compromised subset satisfies the rule. In the benign availability block, hidden authorization is true; affected producers are unavailable; all available producers emit validly signed true claims; and `false_conservative` is true when the rule rejects because insufficient votes or domains remain.

TABLE II
SELECTED STUDY-3 RESIDUAL-BOUNDARY RESULTS

Evidence / contact / policy	Unsafe-qualified trajectories	Mean unsafe-qualified exposure, logical s	Interpretation
Persistent V5, K0, B0	46/46	122.500	Sustained false qualification under continuous contact
Persistent V5, K0, S1	46/46	122.500	Same continuous-contact exposure in frozen grid
Persistent V5, K0, B2	0/46	0	Structural zero in frozen policy/treatment cell
Persistent V5, K4, B0	46/46	55.326	Reduced modeled exposure, not elimination
Persistent V5, K4, S1	46/46	49.022	Additional K4 restriction relative to B0, not immunity
Persistent V5, K4, B2	0/46	0	Structural zero in frozen policy/treatment cell
Truthful V0, K4, B0	3/46	0.326	Pre-onset cache boundary, not adversarial evidence
Truthful V0, K4, S1	0/46	0	No truthful-V0 false qualification in frozen K4 grid
Truthful V0, K4, B2	0/46	0	No truthful-V0 false qualification in frozen K4 grid

Every producer subset is evaluated: 128 subsets per block per rule, for $18 \times 2 \times 128 = 4,608$ exact observations. The two blocks are not combined and do not model simultaneous compromise plus benign loss.

B. First and Systematic Failure Definitions

For each rule and block, the **first failure count** is the smallest affected-producer count for which at least one subset fails; the **systematic failure count** is the smallest count for which every subset fails. The distinction captures subset dependence introduced by provenance composition. Because all subsets are enumerated, these are finite combinatorial properties, not compromise, outage, or mission-availability probabilities.

C. Exact Threshold Map

Table III gives the complete frozen map as first/systematic counts.

Absolute vote count sets the basic compromise boundary while moving benign-loss tolerance in the opposite direction. Provenance can further delay systematic unsafe qualification for selected thresholds, but can also cause earlier false-conservative rejection.

D. Absolute Vote Count Sets the Basic Compromise Boundary

With only one required domain, unsafe qualification follows the vote threshold directly: Q1_D1 through Q7_D1 require one through seven compromised producers. Benign tolerance moves oppositely: Q1_D1 can tolerate six unavailable producers, Q4_D1 fails at 4/4, and Q7_D1 rejects after any single loss. This familiar quorum tradeoff is prior art [7], [8]; Study 4 contributes the exact mapping for the frozen recovery-evidence model and its provenance variants.

E. Provenance Diversity Changes Systematic Failure Without Necessarily Changing First Failure

Q3_D3 shows the clearest provenance effect. First unsafe failure remains at three compromised producers, as under Q3_

D1, but a three-producer subset must span all three domains; systematic failure moves from 3 under Q3_D1 to 6 under Q3_D3. The same pattern appears at Q4_D3 (4/6) and Q5_D3 (5/6). Reporting first and systematic counts together prevents a single threshold from hiding this subset dependence.

F. Provenance Diversity Also Creates Earlier Benign Rejection for Selected Rules

The same structure can reduce benign qualification tolerance. Q3_D1 fails at 5/5 unavailable producers, whereas Q3_D3 first fails at two and becomes systematic at five because two losses can remove an entire domain. Q4_D3 similarly changes benign failure from 4/4 under Q4_D1 to 2/4, and Q5_D3 first fails at two versus three under Q5_D1. These are qualification effects under the frozen denominator and domain assignment, not mission-availability measurements.

G. Null and Equal-Threshold Results

Provenance is not monotonically beneficial. Q4_D1 and Q4_D2 are identical in both blocks at 4/4; Q5_D1 and Q5_D2 are identical at safety 5/5 and benign availability 3/3; all Q6 variants are 6/6 and 2/2; and all Q7 variants are 7/7 and 1/1. The effect of domain requirements is therefore conditional on vote threshold, domain allocation, and subset composition.

H. The Q4 Boundary as a Symmetric Reference Case

Q4_D1 is symmetric at 4/4 in both separately evaluated blocks, whereas Q4_D3 preserves first unsafe failure at four, moves systematic unsafe failure to six, moves first benign failure to two, and leaves systematic benign failure at four. This compact case illustrates why vote count alone does not describe a provenance-constrained rule.

I. High Vote Thresholds and the Loss-Tolerance Boundary

At high thresholds, compromise resistance increases while benign-loss tolerance tightens: Q5_D1, Q6, and Q7 require five, six, and seven compromised producers respectively, but

TABLE III
STUDY-4 FIRST AND SYSTEMATIC FAILURE THRESHOLDS

Rule	Unsafe qualification, compromised producers	False-conservative rejection, unavailable producers
Q1_D1	1/1	7/7
Q2_D1	2/2	6/6
Q2_D2	2/4	4/6
Q3_D1	3/3	5/5
Q3_D2	3/4	4/5
Q3_D3	3/6	2/5
Q4_D1	4/4	4/4
Q4_D2	4/4	4/4
Q4_D3	4/6	2/4
Q5_D1	5/5	3/3
Q5_D2	5/5	3/3
Q5_D3	5/6	2/3
Q6_D1	6/6	2/2
Q6_D2	6/6	2/2
Q6_D3	6/6	2/2
Q7_D1	7/7	1/1
Q7_D2	7/7	1/1
Q7_D3	7/7	1/1

become false-conservative after three, two, and one unavailable producers. With no utility weights, operational probabilities, or mission costs, the grid does not define a globally best rule.

J. Relationship to Distributed Trust Prior Art

Study 4 is a deterministic qualification model, not Byzantine consensus or distributed agreement: producers do not run an agreement protocol, and the model does not analyze leaders, forks, liveness, partitions, replicated state, or Byzantine broadcast. Quorum literature [7], [8] and satellite endorsement-quorum work [9] provide structural prior art; the contribution here is the frozen recovery-qualification threshold map.

K. Study-4 Residual Trust Boundary

Absolute vote count sets the basic compromise boundary, while synthetic provenance constraints change which same-size subsets qualify, delaying systematic failure in selected cells and causing earlier benign rejection in others. The 128 subsets are model states rather than probabilities, the domains are labels rather than demonstrated real independence, and Study 4 contains no contact model. The result is a conditional qualification frontier, not a global policy ranking.

VI. RECOVERY-ARTIFACT ASSURANCE AND RESIDUAL INCORRECT STATES

A. Study Question and Model Boundary

Study 6 (S6-SCTR-001) moves the trust question upstream from runtime authorization evidence to the recovery artifact itself. The study asks when artifact-provenance and release-verification gates can still qualify an objectively incorrect recovery baseline, and what benign qualification cost

is introduced when stronger gates require more assurance signals.

The experiment is an exact finite Boolean assurance model. It does not implement malware, an exploit, a real software-supply-chain compromise, a production build system, real signing keys, or an operational spacecraft recovery pipeline. The artifact states and assurance signals are prespecified model variables. SLSA, TUF, SPARTA, and related assurance concepts motivate the selected dimensions but do not validate the model or establish standards compliance.

The research-only oracle is `objective_baseline_correct`. It identifies whether the modeled recovery artifact is objectively correct, but it is never provided to an assurance gate. A gate sees only its required policy-visible assurance signals.

B. Artifact States and Assurance Signals

The frozen model contains six artifact states. `CLEAN_APPROVED` is objectively correct and has all six visible assurance signals true. Five states are objectively incorrect:

1. `POST_RELEASE_TAMPER`: signature, digest, provenance, and reproduced-build signals are false, while review and approval remain true.
2. `TRUSTED_SIGNER_COMPROMISE`: the signature remains valid, but independent digest, provenance, and reproduced-build signals are false; review is true and release approval is false.
3. `TRUSTED_BUILDER_COMPROMISE`: signature, digest, provenance, review, and approval are true, while independent reproduced-build match is false.
4. `SOURCE_REVIEW_BYPASS`: signature, digest, provenance, reproduced-build, and approval are true, while source-review attestation is false.

5. APPROVED_BAD_SOURCE: all six gate-visible assurance signals are true while objective baseline correctness is false.

The six visible assurance signals are:

- signature_valid;
- independent_target_digest_match;
- provenance_valid;
- independent_reproduced_build_match;
- source_review_attested;
- release_approved.

Terms such as "independent" in the signal names are part of the frozen abstraction. They do not establish that different real organizations, hardware roots, networks, or build infrastructures were used.

C. Assurance Gates and Exact Population

Six deterministic gates compose these signals in progressively different combinations:

- G0_SIGNATURE_ONLY: signature validity;
- G1_SIGNATURE_TARGET_DIGEST: signature plus independent target digest;
- G2_SIGNATURE_PROVENANCE: signature plus provenance;
- G3_PROVENANCE_REPRODUCED_BUILD: signature, provenance, and reproduced-build match;
- G4_PROVENANCE_SOURCE_REVIEW: signature, provenance, and source-review attestation;
- G5_COMPOSITE: all six visible signals.

The experiment has two separate blocks. Block A crosses all six artifact states with all six gates, producing 36 observations. The key endpoint is `unsafe_qualified`, which is true when a gate qualifies an objectively incorrect artifact state.

Block B uses only the objectively correct CLEAN_APPROVED baseline and exhausts all 64 subsets of unavailable assurance signals across all six gates. This produces 384 observations. The endpoint `benign_availability_loss` is true when the gate rejects the correct baseline because one or more required assurance signals are unavailable.

The full frozen population is therefore 420 exact observations. The adversarial artifact-state block and the benign assurance-unavailability block are separate. The study does not model a compromised artifact and assurance-signal loss occurring simultaneously.

D. Signature-Only Qualification Leaves Four Incorrect States

G0_SIGNATURE_ONLY requires only a valid signature. It rejects POST_RELEASE_TAMPER, where the modeled signature is invalid, but qualifies the other four incorrect states: TRUSTED_SIGNER_COMPROMISE, TRUSTED_BUILDER_COMPROMISE, SOURCE_REVIEW_BYPASS, and APPROVED_BAD_SOURCE.

The result is 4 of 5 prespecified incorrect states qualified under the signature-only gate. This ratio is not a detection rate, false-negative rate, or estimate over a real attack population. The five incorrect states are a deliberately constructed finite set, not a sample from software-supply-chain incidents.

The result supports a narrower systems interpretation: signature validity can establish the modeled integrity/authenticity property required by G0, but cannot by itself establish objective recovery-baseline correctness when the signer or upstream production process is inside the modeled trust boundary.

E. Digest and Provenance Close the Signer-Only Gap but Leave Upstream States

G1_SIGNATURE_TARGET_DIGEST adds independent target-digest match. It rejects both POST_RELEASE_TAMPER and TRUSTED_SIGNER_COMPROMISE, reducing unsafe qualification to three states: TRUSTED_BUILDER_COMPROMISE, SOURCE_REVIEW_BYPASS, and APPROVED_BAD_SOURCE.

G2_SIGNATURE_PROVENANCE also qualifies three of the five incorrect states, and in this frozen state set the same three states remain qualified. This equal aggregate count and equal residual set do not establish operational equivalence between independent target-digest verification and provenance verification. The gates represent different assurance mechanisms, and the finite state model does not enumerate every condition under which those mechanisms could diverge.

The relevant result is therefore bounded: either added signal closes the modeled signer-only state while leaving the upstream builder, review, and fully approved bad-source states visible as qualified under this abstraction.

F. Reproduced-Build and Source-Review Gates Are Complementary

G3_PROVENANCE_REPRODUCED_BUILD requires signature validity, provenance validity, and an independent reproduced-build match. It rejects the trusted-builder-compromise state in addition to the post-release-tamper and trusted-signer-compromise states. Two incorrect states remain qualified: SOURCE_REVIEW_BYPASS and APPROVED_BAD_SOURCE.

G4_PROVENANCE_SOURCE_REVIEW also qualifies two of the five incorrect states, but the residual set is different. It qualifies TRUSTED_BUILDER_COMPROMISE and APPROVED_BAD_SOURCE, while rejecting SOURCE_REVIEW_BYPASS.

The equal 2-of-5 counts therefore conceal different trust boundaries. Reproduced-build evidence closes the modeled builder-compromise pathway but does not detect a source-review bypass if the resulting artifact remains reproducible. Source-review evidence closes the modeled review-bypass pathway but does not detect the trusted-builder state when

the required review signal remains true. These gates are complementary in the frozen model rather than interchangeable.

G. Composite Assurance Leaves the Approved-Bad-Source Boundary

G5_COMPOSITE requires all six visible assurance signals. It rejects POST_RELEASE_TAMPER, TRUSTED_SIGNER_COMPROMISE, TRUSTED_BUILDER_COMPROMISE, and SOURCE_REVIEW_BYPASS. The only prespecified incorrect state that remains qualified is APPROVED_BAD_SOURCE.

This state is deliberately important. All six gate-visible signals are true, including signature, independent digest, provenance, reproduced-build match, source-review attestation, and release approval, while the research-only objective-correctness oracle is false. The gate therefore has no visible signal on which to reject the state.

The result should not be overstated. APPROVED_BAD_SOURCE is a structural observability boundary of the frozen model, not an information-theoretic impossibility result. The study does not prove that no assurance mechanism could detect bad source. A different model could include additional evidence, such as independent semantic validation, formal verification, behavioral testing, or other assurance signals. Those signals were not part of the frozen Study-6 design and are not added post hoc to remove the residual state.

The result also overlaps conceptually with established supply-chain guidance. SLSA v1.2 explicitly recognizes that an intentionally malicious software producer cannot be directly mitigated through SLSA controls and that the consumer must establish a basis for trusting the producer [13]. The contribution here is therefore not the discovery of that principle. It is the exact residual-state mapping of the frozen recovery-artifact gate composition.

H. Exact Gate Frontier

Table IV reports the canonical gate summary. The unsafe count is the number of the five prespecified incorrect states that remain qualified. The benign-loss count is the number of the 64 unavailable-signal subsets that cause the objectively correct baseline to be rejected.

For every gate, loss of a single signal that the gate requires is sufficient to produce benign rejection in at least one unavailable-signal subset. The minimum missing-signal count for benign loss is therefore one for all six gates.

Again, the denominators in Table IV are finite model populations. 1/5 must not be interpreted as a 20 percent residual attack rate, and 63/64 must not be interpreted as a 98.4 percent operational outage probability. They describe how many prespecified model states or signal-unavailability subsets cross a deterministic qualification rule.

I. Stronger Gates Increase Sensitivity to Benign Assurance Loss

The benign block exposes the cost of requiring more evidence. G0, which requires only signature validity, rejects the correct baseline in 32 of 64 unavailable-signal subsets. Both two-signal gates reject 48 of 64 subsets. The two three-signal gates reject 56 of 64. The six-signal composite gate rejects 63 of 64, qualifying the correct baseline only when none of its required assurance signals is unavailable.

This monotonic count progression within the frozen gate definitions does not establish a global optimization rule. A stricter gate narrows the modeled set of incorrect states that qualify, but also increases the number of benign missing-evidence states that cause rejection. The study does not assign operational probabilities, mission costs, or utility weights to either side of that frontier.

The term "availability" is therefore used cautiously. Study 6 measures qualification availability under modeled assurance-signal loss. It does not measure mission availability, network availability, spacecraft contact, or service uptime.

J. Relationship to Provenance and Update-Security Prior Art

The Study-6 gates intentionally use assurance concepts that are already established in software-supply-chain and update-security systems. in-toto provides verifiable supply-chain step metadata [10]. TUF uses signed metadata, target hashes, trusted roles, thresholds, and expiration to secure software updates [11]. SLSA defines source assurance requirements and documents supply-chain threat boundaries [12], [13]. SPARTA also motivates integrity-protected and validated recovery baselines in the spacecraft context [4].

Study 6 neither replaces nor validates those systems. It also does not claim standards compliance. The model asks a different question: if a recovery-qualification policy can observe selected assurance signals corresponding to these broad concepts, which prespecified incorrect artifact states remain observationally indistinguishable from acceptable artifacts under each gate?

That question makes the residual state set, rather than the existence of provenance itself, the primary result.

K. Study-6 Residual Trust Boundary

Study 6 shows that composing more artifact-assurance signals closes specific modeled failure pathways, but the residual boundary changes with which assurance dimensions are visible. Signature-only qualification leaves four prespecified incorrect states. Adding digest or provenance closes the signer-only state. Adding reproduced-build evidence closes the modeled builder-compromise state, while adding source-review evidence closes the review-bypass state. The composite gate closes all modeled integrity/provenance pathways except APPROVED_BAD_SOURCE, where every gate-visible assurance signal remains true despite objective incorrectness.

TABLE IV
STUDY-6 RESIDUAL INCORRECT STATES AND BENIGN ASSURANCE LOSS

Gate	Required visible signals	Incorrect states still qualified	Unsafe count	Benign-loss subsets
G0_SIGNATURE_ONLY	1	APPROVED_BAD_SOURCE; SOURCE_REVIEW_BYPASS; TRUSTED_BUILDER_COMPROMISE; TRUSTED_SIGNER_COMPROMISE	4/5	32/64
G1_SIGNATURE_TARGET_DIGEST	2	APPROVED_BAD_SOURCE; SOURCE_REVIEW_BYPASS; TRUSTED_BUILDER_COMPROMISE	3/5	48/64
G2_SIGNATURE_PROVENANCE	2	APPROVED_BAD_SOURCE; SOURCE_REVIEW_BYPASS; TRUSTED_BUILDER_COMPROMISE	3/5	48/64
G3_PROVENANCE_REPRODUCED_BUILD	3	APPROVED_BAD_SOURCE; SOURCE_REVIEW_BYPASS	2/5	56/64
G4_PROVENANCE_SOURCE_REVIEW	3	APPROVED_BAD_SOURCE; TRUSTED_BUILDER_COMPROMISE	2/5	56/64
G5_COMPOSITE	6	APPROVED_BAD_SOURCE	1/5	63/64

The same composition increases sensitivity to benign assurance-signal loss. The result is therefore a finite residual-correctness versus qualification-availability frontier, not a globally best gate. It completes the third study-specific layer needed for the cross-study synthesis: Study 3 addresses temporal runtime evidence, Study 4 addresses producer composition, and Study 6 addresses the recovery artifact itself.

VII. CROSS-STUDY RESIDUAL TRUST BOUNDARIES

A. Scope of the Synthesis

Studies 3, 4, and 6 were designed, executed, and frozen separately. Their populations, mechanisms, and endpoints are not pooled. The synthesis compares only how policy-visible evidence leaves different residual qualification boundaries; it is a manuscript-level interpretation, not a prospectively tested integrated architecture or fourth experiment.

B. Three Qualification Layers

Fig. 1 summarizes the three layers as parallel, separately frozen experiments. The panels do not share a common measurement scale and do not represent experimental data flow or a causal sequence.

The figure preserves the same bounded comparison previously shown in Table V: Study 3 contrasts gate-visible temporal evidence with hidden authorization truth, Study 4 contrasts producer-composition rules with hidden authorization truth, and Study 6 contrasts artifact-assurance signals with objective baseline correctness. The figure is therefore a qualitative mechanism comparison, not a basis for combining numeric outcomes.

C. Integrity Does Not Exhaust Semantic Trust

Study 3 separates post-signature alteration from false content produced inside the modeled trust boundary. V4 invalidates the affected signature and the manipulated record does not qualify. V5 remains validly signed by the trusted producer

and can remain qualified even when hidden authorization truth is false.

Study 6 exposes an analogous upstream boundary. Additional digest, provenance, reproduced-build, review, and approval signals close specific modeled incorrect states, yet APPROVED_BAD_SOURCE remains qualified because every frozen gate-visible signal is true while objective correctness is false.

The implication is bounded: integrity, freshness, provenance, and process evidence establish only the properties represented by those signals and their trust anchors. They do not automatically reveal a semantic mismatch that the gate cannot observe.

D. Stronger Composition Moves the Boundary

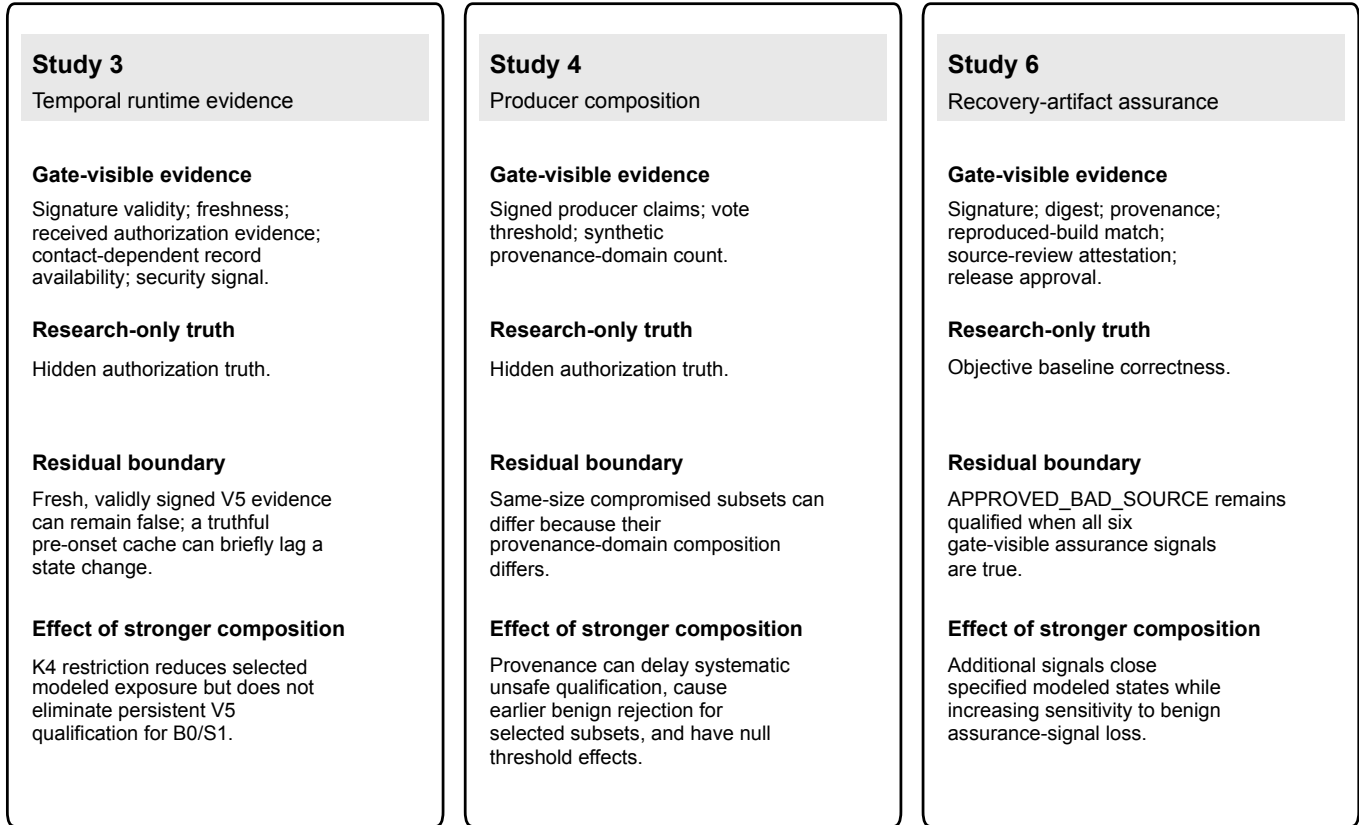
Study 4 shows that additional provenance structure can delay systematic unsafe qualification without always changing first failure. For example, Q3_D3 leaves first unsafe failure at three compromised producers but moves systematic failure from three under Q3_D1 to six. The same constraint makes benign false-conservative rejection possible after two unavailable producers rather than five. Other provenance additions produce no threshold change, so diversity is not monotonically beneficial in the frozen model.

Study 6 shows a different frontier. Stronger gates reduce the prespecified incorrect states that remain qualified from four under signature-only checking to one under the six-signal composite gate, while benign-loss subsets increase from 32/64 to 63/64. Equal counts can still hide different residual mechanisms, as G3 and G4 each leave two incorrect states but not the same two.

Study 3 is not folded into that availability frontier because it has different endpoints. Its contact-aware restriction reduces selected K4 exposure while persistent V5 qualification remains present for B0 and S1. Across all three studies, stronger composition changes a boundary condition rather than establishing universal dominance.

Three separately frozen experiments | qualitative synthesis only

No pooled population | no experimental data flow between panels



Parallel panels are a qualitative manuscript synthesis, not an integrated recovery architecture. Only Study 3 models contact.

Fig. 1. Parallel residual trust boundaries across the three separately frozen studies. The panels summarize a qualitative manuscript-level synthesis only; no experimental data flow or integrated three-layer architecture connects Studies 3, 4, and 6. Only Study 3 models contact.

E. Residual Identity Matters

Aggregate count or duration is insufficient to identify the remaining trust assumption. Study 4 distinguishes first from systematic failure because same-size producer subsets can differ in provenance composition. Study 6 preserves residual state identity because gates with equal unsafe counts can fail on different artifact states. Study 3 preserves false-qualification origin because a truthful pre-onset cache and a compromised-producer record represent different mechanisms.

This is why the manuscript reports origin, subset structure, and surviving artifact states rather than collapsing the studies into one scalar trust score.

F. Observability as the Common Constraint

The common principle is observability. Study 3 cannot directly observe that a trusted signer is semantically lying; Study 4 cannot observe a compromise oracle beyond the

signed claims and structural provenance labels supplied to the rule; Study 6 cannot observe objective correctness when every required assurance signal remains true.

Within each frozen model, changing the arrangement or quantity of visible evidence can narrow the set of qualifying failures, but it cannot discriminate a mismatch that remains observationally identical under the gate's variables. This is a model-specific systems result, not a universal impossibility theorem.

G. Aerospace Systems Implications

The experiments suggest four design questions for aerospace information systems. First, recovery requirements should distinguish evidence integrity from authority and semantic trust. Second, multi-source evidence should document what operational failure separation a claimed provenance domain represents. Third, stronger evidence requirements should be assessed together with the benign conditions under which required evidence can become unavailable. Fourth, artifact

assurance should identify the highest-level trust assumption that remains outside the gate.

These implications are not prescriptive flight requirements. Mission-specific adoption would require mapping the abstract producers, provenance domains, timing semantics, gates, and failure states to an actual architecture and validating that mapping under operational conditions.

H. Synthesis Result

Taken together, the studies support a layered residual-trust interpretation of satellite cyber-recovery qualification. Temporal evidence, producer composition, and artifact assurance each close some modeled failure pathways while leaving a different residual assumption outside direct observation. The synthesis therefore does not identify a globally best policy, producer-composition rule, or artifact gate. Its contribution is to make the remaining trust assumption explicit at each layer without pooling the three experiments.

VIII. VALIDITY, AEROSPACE INTERPRETATION BOUNDARIES, AND FUTURE EVALUATION

A. Internal and Construct Validity

The three experiments are exact evaluations of frozen finite models, so their strongest validity claim is internal to those models. Study 3 preserves the distinction between policy-visible evidence and hidden authorization truth, prespecifies the two allowable origins of false qualification, and independently audits the resulting trajectories and epoch rules. Study 4 exhausts every producer subset for every registered quorum/provenance rule in two separately defined blocks. Study 6 exhausts its prespecified artifact states and all subsets of benign assurance-signal unavailability. These controls reduce ambiguity about what each frozen endpoint represents.

The common residual-trust framework introduced in this manuscript was not itself a prospectively tested experimental treatment. Studies 3, 4, and 6 were designed and frozen independently. The common framework is a post hoc systems-level interpretation used to compare their mechanisms without changing the frozen outcomes. It should therefore be evaluated as a synthesis of three separately supported results, not as evidence that an integrated three-layer recovery architecture was experimentally validated.

The endpoint terminology also requires care. In Study 3, `unsafe_qualified` means that the modeled recovery gate remains policy-visible qualified while hidden authorization truth is false. In Study 4, the safety block measures resistance to unsafe qualification, and the availability block measures false-conservative rejection under benign producer unavailability. In Study 6, benign availability loss means rejection of the objectively correct artifact when required assurance signals are unavailable. None of these constructs is equivalent to spacecraft physical safety, mission availability, or successful completion of a recovery procedure.

B. Study-3 Boundary Conditions

Study 3 uses one continuous-contact regime and one synthetic intermittent-contact regime. K4 contains four fixed contact windows over a 240-logical-second horizon. The result therefore characterizes that registered schedule and the associated five-logical-second epoch and freshness semantics. It does not establish how the same policies would behave under other contact schedules, real orbital geometry, different latency distributions, variable evidence lifetimes, or mission-specific ground coverage.

Logical seconds are model units. Although the labels preserve the frozen timing semantics, they are not measurements of spacecraft processor time, radio-link latency, ground-station delay, operator response time, or elapsed orbital time. The reported 122.5, 55.326, 49.022, 5, and 0.326 logical-second exposures must therefore remain model quantities.

The 46 onset phases exhaust the frozen onset grid, but they are not 46 random draws from an operational distribution. Similarly, the 67,620 epoch states are repeated states nested within 1,380 trajectories and are not independent statistical observations. The trajectory remains the study unit.

The trusted-producer compromise in V5 is also abstract. It assumes that a trusted producer can validly sign a false authorization claim. The experiment does not model how that compromise occurs, whether a signing key is stolen, whether software is maliciously modified, or how likely the condition is in a real mission. Conversely, V4 models a post-signature value change that invalidates the signature. The contrast establishes a bounded difference between altered signed data and false data signed by the trusted producer; it does not evaluate cryptographic strength or key-management security.

C. Study-4 Boundary Conditions

Study 4 fixes the producer population at seven and the synthetic provenance allocation at 3/2/2. Its exact thresholds are therefore conditional on that registered producer set, domain allocation, denominator, and 18 rule definitions. A different number of producers, a different provenance allocation, dynamic membership, weighted voting, or a responder-based denominator could produce different thresholds.

The provenance domains are synthetic independence classes. The experiment does not demonstrate organizational, hardware, software, network, administrative, sensing-path, or supply-chain independence among real producers. Any operational use of a provenance constraint would require an external justification for what failure separation the domains represent.

The safety and benign-availability populations are deliberately separate. The study does not evaluate simultaneous malicious compromise and benign producer unavailability. It also does not model adaptive adversaries, collusion strategies beyond the affected-subset state, network timing, Byzantine agreement, leader election, or sensor-estimation error.

The 128 subsets per block are the complete power set of the seven modeled producers. Fractions of failing subsets at a given affected-producer count are therefore combinatorial properties of the model, not probabilities that a real subset will be compromised or unavailable. First and systematic failure thresholds likewise describe the frozen set structure and should not be interpreted as operational reliability limits.

D. Study-6 Boundary Conditions

Study 6 is an abstract six-state, six-gate Boolean assurance model. The five objectively incorrect states were selected to expose distinct trust assumptions, but they are not an exhaustive taxonomy of software-supply-chain failure. Consequently, values such as 4/5, 3/5, 2/5, and 1/5 are finite state counts rather than detection rates, false-negative rates, or estimates of residual compromise probability.

The six assurance signals are also modeled Boolean variables. `independent_target_digest_match` and `independent_reproduced_build_match` do not demonstrate real organizational or infrastructure independence. `source_review_attested` and `release_approved` do not measure the quality, competence, or adversarial resistance of a real review and approval process.

`APPROVED_BAD_SOURCE` is intentionally defined so that all six gate-visible signals are true while objective correctness is false. It therefore identifies the observability boundary of the frozen model. The result is not a theorem that semantic correctness can never be established, nor does it imply that additional techniques such as formal verification, independent behavioral testing, semantic review, runtime validation, or diverse implementations could not add evidence. Those mechanisms were outside the frozen Study-6 design and were not added after observing the result.

The benign assurance-unavailability block is likewise structural. All 64 unavailable-signal subsets are evaluated with the objectively correct baseline. Counts such as 63/64 describe deterministic rejection across that finite subset space. They do not estimate service outage probability, contact probability, or the frequency with which real assurance systems become unavailable.

E. External Validity and Aerospace Generalization

No experiment in Paper 2 operates an on-orbit spacecraft, ground station, RF link, flight processor, production key infrastructure, or real mission command path. No operational spacecraft recovery is executed. The results therefore do not establish flightworthiness, certification, mission assurance compliance, operational attack prevalence, recovery success probability, or mission-level availability.

Only Study 3 directly includes a contact variable. Study 4 producer unavailability must not be interpreted as loss of spacecraft contact, and Study 6 assurance-signal unavailability

must not be interpreted as either contact loss or network outage. Treating all three as manifestations of intermittent connectivity would erase important construct differences among the experiments.

The spacecraft relevance instead comes from the systems question being modeled: recovery qualification can depend on evidence received under intermittent contact, evidence composed across trusted producers, and assurance about the recovery artifact itself. The experiments isolate those mechanisms in deterministic abstractions. Generalization to a specific spacecraft program would require mapping the abstract evidence producers, provenance domains, gates, timing semantics, and failure states to mission-specific components and then validating that mapping against the target architecture.

F. Statistical Interpretation

The studies evaluate complete finite populations specified by their frozen protocols. Sampling-based inferential statistics are therefore not used to make claims about the registered grids. No p-value gate, confidence interval over an assumed superpopulation, or pooled effect estimate is required to establish an exact count or threshold within the enumerated state space.

This does not make the results universal. Exactness applies to the registered model population, not to all possible spacecraft or attack conditions. The absence of sampling uncertainty within a finite grid is distinct from uncertainty about model choice, external validity, or omitted operational variables.

The three populations also remain incommensurate. Study 3 uses trajectories, Study 4 uses rule-by-subset observations, and Study 6 uses artifact-state and assurance-unavailability observations. Their arithmetic sum is not a scientifically meaningful sample size. No pooled N, success percentage, confidence interval, or global rank is defined in this paper.

G. Reproducibility and Independence

Each experiment is provenance-bound and independently audited within the repository. Study 3 reports zero trajectory, epoch-rule, false-qualification-origin, and hash mismatches. Study 4 reports zero observation and threshold mismatches under independent reconstruction. Study 6 reports zero mismatches with matching frozen outputs and no tracked-file drift.

These controls support same-repository reproducibility and strengthen confidence that the reported manuscript projections reflect the frozen experiments. They do not constitute external empirical replication because the independent implementations and audits remain within the same research program and repository. External replication would require an independent research group, environment, or evidence source beyond the present repository controls.

H. Standards and Framework Interpretation

RATS, SPARTA, SLSA, TUF, in-toto, and related sources are used to position the modeled evidence dimensions against established security concepts. The studies do not implement every requirement of those frameworks, and no compliance assessment was performed. References to provenance, attestation, reproducible builds, trusted baselines, or cyber-safe recovery therefore identify conceptual relationships only.

This distinction is particularly important for Study 6. The presence of model variables named provenance, reproduced build, source review, and release approval does not establish that a deployed implementation would satisfy SLSA, TUF, SPARTA, or any other standard or framework.

I. Limits of the Cross-Study Synthesis

The cross-study synthesis compares residual mechanisms, not causal transitions from one experiment to the next. Study 4 does not experimentally mitigate Study 3, and Study 6 does not experimentally validate a downstream artifact gate for either Study 3 or Study 4. No data flow connects the frozen populations.

The synthesis also does not imply that the three qualification layers are complete. Real recovery systems can depend on additional dimensions, including command authority, hardware roots of trust, behavioral verification, physical-state estimation, network path integrity, human authorization, fault-management logic, and mission-phase constraints. The present paper focuses only on the three independently frozen mechanisms that were evaluated.

Accordingly, the layered residual-trust interpretation should be read as an analytical decomposition: temporal evidence, producer composition, and artifact assurance each expose a distinct trust boundary. It is not a claim that every recovery architecture should contain exactly these three layers or evaluate them in this order.

J. Future Evaluation

Several extensions could test the portability of the present findings without altering the frozen evidence reported here. Study 3 could be complemented by prospectively designed orbital-contact schedules, variable evidence lifetimes, or hardware/software-in-the-loop timing measurements. Study 4 could be extended to different producer counts, empirically justified failure domains, dynamic membership, or joint compromise-and-unavailability conditions. Study 6 could be evaluated against real build pipelines, independently operated assurance services, additional semantic-validation mechanisms, or prospectively defined joint artifact-compromise and evidence-loss scenarios.

A separate integrated experiment could also test how run-time evidence, multi-producer composition, and artifact assurance interact in one recovery architecture. Such an experiment would be scientifically different from the present manuscript

because it would define new joint interventions, units, and endpoints. It should therefore be designed and frozen prospectively rather than inferred from the three existing populations.

These extensions are opportunities for external validation and broader generalization. They are not defects that justify reopening or rerunning the frozen Studies 3, 4, or 6 for the present paper.

IX. CONCLUSION

Paper 2 characterizes trusted cyber-recovery qualification through three separately frozen deterministic studies rather than one integrated experiment. Study 3 separates brief truthful pre-onset cache exposure from false but validly signed $\nabla 5$ evidence issued by a compromised trusted producer; persistent $\nabla 5$ remains qualified for B0 and S1 across all 46 onset phases under both $\kappa 0$ and synthetic $\kappa 4$, although $\kappa 4$ reduces modeled exposure. Affected $\nabla 4$ post-signature-manipulated records have invalid signatures and do not qualify.

Study 4 shows that absolute vote count establishes the basic compromise boundary while synthetic provenance-domain requirements change which same-size subsets can satisfy it. Selected requirements delay systematic unsafe qualification but cause earlier benign false-conservative rejection, while other domain requirements have null threshold effects. Study 6 similarly shows that composed artifact-assurance signals close specific incorrect states: the six-signal gate leaves only `APPROVED_BAD_SOURCE`, while stronger gates increase rejection under benign assurance-signal loss.

Across the three studies, stronger trust composition moves or narrows modeled qualification boundaries without making policy-visible evidence equivalent to hidden or objective truth. Residual identity matters as much as aggregate count or duration. The results remain bounded to finite models: only Study 3 models contact, logical time is not operational spacecraft time, synthetic domains do not establish real independence, Study 6 is not a real supply-chain experiment, and no pooled population, global policy ranking, flight-safety claim, mission-availability claim, or operational recovery probability is inferred.

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REFERENCES

- [1] R. Thummala, E. Rice, and G. Falco, "Why is space cybersecurity unique?," in Proc. 4th Workshop Security Space Satellite Syst. (SpaceSec), San Diego, CA, USA, Feb. 23, 2026, doi: 10.14722/spacesec.2026.23055.
- [2] J. Vanlyssel, G.-C. Roman, K. Cook, S. Rahaman, and A. Anwar, "Trust without boundaries: An architectural analysis of satellite flight software," 2026, arXiv:2608.14532.
- [3] J. Curbo and G. Falco, "Testable cyber requirements for space flight software," in Proc. 2025 IEEE Aerosp. Conf., Big Sky, MT, USA, 2025, pp. 1-20, doi: 10.1109/AERO63441.2025.11068629.
- [4] The Aerospace Corporation. "Space Attack Research & Tactic Analysis (SPARTA)." Accessed: Sep. 6, 2026. [Online]. Available: <https://sparta.aerospace.org/>
- [5] H. Birkholz, D. Thaler, M. Richardson, N. Smith, and W. Pan, "Remote Attestation procedureS (RATS) architecture," RFC 9334, Jan. 2023, doi: 10.17487/RFC9334.
- [6] Y. Deshpande, J. Zhang, H. Labiod, and H. Birkholz, "Remote attestation with multiple verifiers," IETF, Internet-Draft draft-ietf-rats-multi-verifier-00, May 2026, work in progress. [Online]. Available: <https://datatracker.ietf.org/doc/draft-ietf-rats-multi-verifier/>
- [7] D. Malkhi and M. Reiter, "Byzantine quorum systems," Distrib. Comput., vol. 11, no. 4, pp. 203-213, Oct. 1998, doi: 10.1007/s004460050050.
- [8] O. Alpos, C. Cachin, B. Tackmann, and L. Zanolini, "Asymmetric distributed trust," Distrib. Comput., vol. 37, no. 3, pp. 247-277, May 2024, doi: 10.1007/s00446-024-00469-1.
- [9] F. Rezabek, D. Malkhi, and A. Yahalom, "Space Fabric: A satellite-enhanced trusted execution architecture," 2026, arXiv:2603.23745.
- [10] S. Torres-Arias, H. Afzali, T. K. Kuppusamy, R. Curtmola, and J. Cappel, "in-toto: Providing farm-to-table guarantees for bits and bytes," in Proc. 28th USENIX Security Symp. (USENIX Security 19), Santa Clara, CA, USA, Aug. 2019, pp. 1393-1410.
- [11] The Update Framework. "The Update Framework Specification, v1.0.36." Aug. 10, 2026. [Online]. Available: <https://github.com/theupdateframework/specification/releases/tag/v1.0.36>
- [12] SLSA. "Source: Requirements for producing source," SLSA Specification, v1.2. Accessed: Sep. 6, 2026. [Online]. Available: <https://slsa.dev/spec/v1.2/source-requirements>
- [13] SLSA. "Threats & mitigations," SLSA Specification, v1.2. Accessed: Sep. 6, 2026. [Online]. Available: <https://slsa.dev/spec/v1.2/threats>